

■ 시나리오 별 백업 복구(User Managed)

KEY WORDS

- <시나리오 3> 백업 받지 않은 테이블스페이스의 데이터파일이 손상(리두정보 O)(drop tablespace X)
- <시나리오 4> 백업 받지 않은 테이블스페이스의 데이터파일이 손상(리두정보 X)(drop tablespace X)
- <시나리오 5> system 데이터파일 손상(백업이후 리두정보 O)
- <시나리오 6> system 데이터파일 손상(백업이후 리두정보 X)
- <시나리오 7> undo datafile 손상(백업 이후 리두정보 O)
- <시나리오 8> undo datafile 손상(백업 이후 리두정보 X)

<<시나리오 3>> 백업 받지 않은 테이블스페이스의 데이터파일이 손상(리두정보 O)(drop tablespace X)

SQL> set linesize 160

SQL> col file_name for a60

SQL> select tablespace_name, file_name from dba_data_files

TABLESPACE_NAME	FILE_NAME
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf

SQL> create tablespace tbs datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' size 10m;
Tablespace created.

SQL> select tablespace_name, file_name from dba_data_files

TABLESPACE_NAME	FILE_NAME
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
TBS	/u01/app/oracle/oradata/ORCL/tbs01.을

SYS@orcl> create table hr.emp1(id number) tablespace tbs;

Table created.

SYS@orcl> insert into hr.emp1 values(1);

1 row created.

SYS@orcl> commit;

Commit complete.

■ 장애 발생

SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf

-rw-r-----. 1 oracle dba 10493952 May 2 19:24 /u01/app/oracle/oradata/ORCL/tbs01.dbf

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/tbs01.dbf

SYS@orcl> col name for a70

SYS@orcl> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2063691	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2064668	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2063691	ONLINE

SYS@orcl> select * from v\$log;

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#
FIRST_TIME	NEXT_CHANGE#	NEXT_TIME						
CON_ID								
1	1	55	20971520	512	1	NO	INACTIVE	2055258 02-MAY-25
2062476	02-MAY-25							
0								
2	1	56	20971520	512	1	NO	CURRENT	2062476 02-MAY-25
1.8447E+19								
0								
3	1	54	20971520	512	1	NO	INACTIVE	2047231 01-MAY-25
2055258	02-MAY-25							
0								

SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' offline drop;

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2063691	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2064668	RECOVER
/u01/app/oracle/oradata/ORCL/users01.dbf	2063691	ONLINE

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

- 백업본이 없는 손상된 데이터파일에 대해서 물리적인 파일을 생성

```
SYS@orcl> alter database create datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
```

```
Database altered.
```

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
-rw-r-----. 1 oracle dba 10493952 May  2 19:45 /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

- 새롭게 생성된 파일에 리두정보를 이용해서 복구 작업

```
SYS@orcl> alter database recover datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
```

```
Database altered.
```

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2063691	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2067356	OFFLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2063691	ONLINE

- 복구 수행한 데이터파일을 online으로 변경

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' online;
```

```
Database altered.
```

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS

/u01/app/oracle/oradata/ORCL/system01.dbf	2063691	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2063691	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2067633	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2063691	ONLINE

```
SYS@orcl> select * from hr.emp1;
```

ID

1

■ 복구 끝난 테이블 스페이스 삭제

```
SYS@orcl> drop tablespace tbs including contents and datafiles;
```

Tablespace dropped.

<<시나리오 4>> 백업 받지 않은 테이블스페이스의 데이터파일이 손상(리두정보 X)(drop tablespace X)

```
SYS@orcl> create tablespace tbs datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' size 10m;
```

```
Tablespace created.
```

```
SYS@orcl> select tablespace_name, file_name from dba_data_files;
```

TABLESPACE_NAME	FILE_NAME
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
SYSaux	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
TBS	/u01/app/oracle/oradata/ORCL/tbs01.dbf

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#	
FIRST_TIME	NEXT_CHANGE#	NEXT_TIME							
CON_ID									
1	1	55	20971520	512	1	NO	INACTIVE	2055258	02-MAY-25
	2062476	02-MAY-25							
2	1	56	20971520	512	1	NO	CURRENT	2062476	02-MAY-25
	1.8447E+19								
3	1	54	20971520	512	1	NO	INACTIVE	2047231	01-MAY-25
	2055258	02-MAY-25							

```
SYS@orcl> create table hr.emp1 tablespace tbs as select * from hr.employees;
```

```
Table created.
```

```
SYS@orcl> select count(*) from hr.emp1;
```

```
COUNT(*)
```

```
107
```

```
SYS@orcl> alter system switch logfile;
```

```
SYS@orcl> /
```

SYS@orcl> /

System altered.

SYS@orcl> select * from v\$log;

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#	
FIRST_TIME	NEXT_CHANGE#		NEXT_TIME						

CON_ID									

1	1	58	20971520	512	1	NO	ACTIVE	2068343	02-
MAY-25	2069059		02-MAY-25						
0									
2	1	59	20971520	512	1	NO	CURRENT	2069059	02-
MAY-25	1.8447E+19								
0									
3	1	57	20971520	512	1	NO	ACTIVE	2068340	02-
MAY-25	2068343		02-MAY-25						
0									

SYS@orcl> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS

/u01/app/oracle/oradata/ORCL/system01.dbf	2068340	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2068340	ONLINE

SYS@orcl> select name dbname, open_mode, log_mode, checkpoint_change# from v\$database;

DBNAME	OPEN_MODE	LOG_MODE	CHECKPOINT_CHANGE#

ORCL	READ WRITE	NOARCHIVELOG	2068340

■ 장애 발생

SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf

-rw-r-----. 1 oracle dba 10493952 May 2 20:05 /u01/app/oracle/oradata/ORCL/tbs01.dbf

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/tbs01.윌

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2068340	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2068340	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2068340	ONLINE

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' offline drop;
Database altered.
```

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2068343	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2068343	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2068343	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2068340	RECOVER
/u01/app/oracle/oradata/ORCL/users01.dbf	2068343	ONLINE

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

- 백업본이 없는 손상된 데이터파일에 대해서 물리적인 파일을 생성

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

```
SYS@orcl> alter database create datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' ;
Database altered.
```

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
-rw-r-----, 1 oracle dba 10493952 May  2 20:13 /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

- 새롭게 생성된 파일에 리두정보를 이용해서 복구 작업을 수행하려고 시도 했지만 리두정보가 없어서 오류발생 복구 실패

```
SYS@ora19c> alter database recover datafile '/u01/app/oracle/oradata/ORA19C/hr_tbs.dbf';
alter database recover datafile '/u01/app/oracle/oradata/ORA19C/hr_tbs.dbf'
*
```

```
ERROR at line 1:
```

```
ORA-00279: change 3274774 generated at 02/13/2025 11:01:27 needed for thread 1
```

```
ORA-00289: suggestion : /u01/app/oracle/fast_recovery_area/ORA19C/archivelog/2025_02_13/o1_mf_1_270_%u_arc
```

ORA-00280: change 3274774 for thread 1 is in sequence #270

SYS@ora19c> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORA19C/system01.dbf	3275012	SYSTEM
/u01/app/oracle/oradata/ORA19C/undotbs01.dbf	3275012	ONLINE
/u01/app/oracle/oradata/ORA19C/sysaux01.dbf	3275012	ONLINE
/u01/app/oracle/oradata/ORA19C/hr_tbs.dbf	3275012	RECOVER
/u01/app/oracle/oradata/ORA19C/users01.dbf	3275012	ONLINE

- 손상된 데이터파일에 대해서는 online으로 변경할 수 없다.

SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' online;

alter database datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' online

*

ERROR at line 1:

ORA-01113: file 5 needs media recovery

ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

- 손상된 데이터 파일이 있는 테이블스페이스를 삭제할 수 없다.

SYS@orcl> drop tablespace tbs including contents and datafiles;

drop tablespace tbs including contents and datafiles

*

ERROR at line 1:

ORA-01156: recovery or flashback in progress may need access to files

- 재시작 후 drop문장 사용 가능

SYS@orcl> shutdown immediate

SYS@orcl> startup

SYS@orcl> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2069626	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2069626	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2069626	ONLINE
/u01/app/oracle/oradata/ORCL/tbs01.dbf	2068340	RECOVER
/u01/app/oracle/oradata/ORCL/users01.dbf	2069626	ONLINE

SYS@orcl> drop tablespace tbs including contents and datafiles;

Tablespace dropped.

<<시나리오 5>> system 데이터파일 손상(백업이후 리두정보 O)

```
SYS@orcl> select name from v$datafile union
2 select name from v$controlfile union
3 select member from v$logfile;
```

NAME

```
/u01/app/oracle/oradata/ORCL/control01.ctl
/u01/app/oracle/oradata/ORCL/control02.ctl
/u01/app/oracle/oradata/ORCL/redo01.log
/u01/app/oracle/oradata/ORCL/redo02.log
/u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
/u01/app/oracle/oradata/ORCL/system01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/users01.dbf
9 rows selected.
```

```
SYS@orcl> create pfile='/home/oracle/backup/noarch/initorcl.ora' from spfile;
File created.
```

```
SYS@orcl> shut immediate
```

```
SYS@orcl> !
```

```
[oracle@oel7v9r2 ~]$ cd backup/noarch/
```

```
[oracle@oel7v9r2 noarch]$ ls
```

```
initorcl.ora
```

```
[oracle@oel7v9r2 noarch]$ cp -v /u01/app/oracle/oradata/ORCL/*.{ctl,log,dbf} .
```

```
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> './control01.ctl'
```

```
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> './control02.ctl'
```

```
'/u01/app/oracle/oradata/ORCL/redo01.log' -> './redo01.log'
```

```
'/u01/app/oracle/oradata/ORCL/redo02.log' -> './redo02.log'
```

```
'/u01/app/oracle/oradata/ORCL/redo03.log' -> './redo03.log'
```

```
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> './sysaux01.dbf'
```

```
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> './system01.dbf'
```

```
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> './temp01.dbf'
```

```
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> './undotbs01.dbf'
```

```
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> './users01.dbf'
```

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME

CHECKPOINT_CHANGE# STATUS

```
/u01/app/oracle/oradata/ORCL/system01.dbf      2072112 SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf      2072112 ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf     2072112 ONLINE
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf          2072112 ONLINE
/u01/app/oracle/oradata/ORA19C/users01.dbf        3266075 ONLINE
```

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#	FIRST_TIME	NEXT_CHANGE#	NEXT_TIME	CON_ID
1	1	58	20971520	512	1	NO	INACTIVE	2068343	02-MAY-25	2069059	02-MAY-25	0
2	1	59	20971520	512	1	NO	CURRENT	2069059	02-MAY-25	1.8447E+19		0
3	1	57	20971520	512	1	NO	INACTIVE	2068340	02-MAY-25	2068343	02-MAY-25	0

■ 장애 발생

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/system01.dbf
-rw-r-----, 1 oracle dba 943726592 May  2 20:43 /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/system01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/system01.dbf: No such file or directory
```

```
SYS@orcl> alter system checkpoint;
```

```
alter system checkpoint
```

```
*
```

```
ERROR at line 1:
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 12555
```

```
Session ID: 621 Serial number: 11641
```

```
SYS@orcl> select status from v$instance;
```

```
ERROR:
```

```
ORA-03114: not connected to ORACLE
```

```
SYS@orcl> conn /as sysdba
```

```
Connected to an idle instance.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1811936368 bytes
```

```
Fixed Size                  8897648 bytes
```

Variable Size 436207616 bytes

Database Buffers 1358954496 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' -- open 시점에 오류 발생

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME
CON_ID				

1	ONLINE	ONLINE	FILE NOT FOUND	0

SYS@orcl> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS

/u01/app/oracle/oradata/ORCL/system01.dbf	2072112	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2072112	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2072112	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2072112	ONLINE

SYS@orcl> select status from v\$instance;

STATUS

MOUNTED

- system data file이 손상되었을 경우 mount 단계에서 복구작업 진행

-- system, current undo, temp datafile은 offline으로 변경이 불가능하다.

- 가장 최근의 백업 받은 system datafile을 찾아서 restore

SYS@orcl> !pwd

/home/oracle/backup/noarch

SYS@orcl> !cp -v system01.dbf /u01/app/oracle/oradata/ORCL/
'system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'

- 백업 이후에 변경정보(리두)를 이용해서 복구작업

SYS@orcl> recover tablespace system;

Media recovery complete.

SYS@orcl> alter database open;

<<시나리오 6>> system 데이터파일 손상(백업이후 리두정보 X)

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2274379	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2274379	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2274379	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2274379	ONLINE

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#	
FIRST_TIME	NEXT_CHANGE#	NEXT_TIME	CON_ID						
1	1	58	20971520	512	1	NO	INACTIVE	2068343	
MAY-25	2069059	02-MAY-25		0					
2	1	59	20971520	512	1	NO	INACTIVE	2069059	
MAY-25	2273158	02-MAY-25		0					
3	1	60	20971520	512	1	NO	CURRENT	2273158	
MAY-25	1.8447E+19			0					

```
SYS@orcl> alter system switch logfile;
```

```
SYS@orcl> /
```

```
SYS@orcl> /
```

■ 장애 발생

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
-rw-r-----. 1 oracle dba 943726592 May  2 20:59 /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> !ls -l /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/system01.dbf: No such file or directory
```

```
SYS@orcl> alter system checkpoint;
```

```
System altered.
```

```
SYS@orcl> conn hr/hr
```

```
ERROR:
```

```
ORA-00604: error occurred at recursive SQL level 1
```

```
ORA-01116: error in opening database file 1
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file directory
Additional information: 3
ORA-00604: error occurred at recursive SQL level 1
ORA-01116: error in opening database file 1
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

Warning: You are no longer connected to ORACLE.

@>

@> conn /as sysdba

Connected.

SYS@orcl> shut abort

ORACLE instance shut down.

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1811936368 bytes

Fixed Size 8897648 bytes

Variable Size 436207616 bytes

Database Buffers 1358954496 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_	ERROR	CHANGE#	TIME
CON_ID					
1	ONLINE	ONLINE	FILE NOT FOUND		0

SYS@orcl> select name, checkpoint_change#, status from v\$datafile;

NAME	CHECKPOINT_CHANGE#	STATUS
/u01/app/oracle/oradata/ORCL/system01.dbf	2275293	SYSTEM
/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2275293	ONLINE
/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2275293	ONLINE
/u01/app/oracle/oradata/ORCL/users01.dbf	2275293	ONLINE

■ system data file이 손상되었을 경우 mount 단계에서 복구작업 진행
-- system, current undo, temp datafile은 offline으로 변경이 불가능하다.

■ 가장 최근의 백업 받은 system datafile을 찾아서 restore
SYS@orcl> !pwd
/home/oracle/backup/noarch

SYS@orcl> !cp -v system01.dbf /u01/app/oracle/oradata/ORCL/
'system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'

■ 백업 이후에 변경정보(리두)를 이용해서 복구작업하려고 시도했지만 복구 실패(완전복구 실패)

SYS@orcl> recover datafile 1;
ORA-00279: change 2072109 generated at 05/02/2025 20:29:32 needed for thread 1
ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_59_1199981728.dbf
ORA-00280: change 2072109 for thread 1 is in sequence #59

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_59_1199981728.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_59_1199981728.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

■ 백업본을 통째로 부어 이전 버전으로 복구(불완전 복구)

SYS@orcl> shut abort
ORACLE instance shut down.
SYS@orcl>
SYS@orcl> !
[oracle@oel7v9r2 noarch]\$ ls
control01.ctl initorcl.ora redo02.log sysaux01.dbf temp01.dbf users01.dbf
control02.ctl redo01.log redo03.log system01.dbf undotbs01.dbf
[oracle@oel7v9r2 noarch]\$ cp -v * /u01/app/oracle/oradata/ORCL/
'control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
'initorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initorcl.ora'
'redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'

```
'redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'  
'sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> startup  
ORACLE instance started.
```

```
Total System Global Area 1811936368 bytes  
Fixed Size                  8897648 bytes  
Variable Size              436207616 bytes  
Database Buffers          1358954496 bytes  
Redo Buffers               7876608 bytes  
Database mounted.  
Database opened.
```

<<시나리오7>> undo datafile 손상(백업 이후 리두정보 O)

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2102516	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2102516	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2102516	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2102516	ONLINE

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

```
SYS@orcl> select s.username, s.sid, s.serial#, r.name, t.ubafil, t.xidusn, t.ubablk, t.used_ublk from v$session s, v$transaction t,
v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

no rows selected

<<hr>>

```
HR@oracl> select salary from hr.employees where employee_id = 100;
```

SALARY
24000

```
HR@orcl> update hr.employees set salary = 30000
where employee_id = 100;
```



```
HR@orcl> select salary from hr.employees where employee_id = 100;
```

```
SALARY
```

```
-----
```

```
30000
```

```
<<sys>>
```

```
SYS@orcl> select s.username, s.sid, s.serial#, r.name, t.ubafil, t.xidusn, t.ubablk, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

USERNAME					SID	SERIAL#

NAME	UBAFIL	XIDUSN	UBABLK	USED_UBLK		

SYS					33	32757
_SYSSMU4_625702278\$	4	4	182	1		

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory
```

```
SYS@orcl> alter system checkpoint;
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> conn /as sysdba
```

```
Connected.
```

```
SYS@orcl> shut abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 4 - see DBWR trace file

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME
CON_ID				

4	ONLINE	ONLINE	FILE NOT FOUND	0
0				

SYS@orcl> select name dbname, open_mode, log_mode, checkpoint_change# from v\$database;

DBNAME	OPEN_MODE	LOG_MODE	CHECKPOINT_CHANGE#

ORCL	MOUNTED	NOARCHIVELOG	2104850

SYS@orcl> select file#, name, checkpoint_change#, status from v\$datafile;

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
-----	-----	-----	-----
1	/u01/app/oracle/oradata/ORCL/system01.dbf	2104850	SYSTEM
3	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2104850	ONLINE
4	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2104850	ONLINE
7	/u01/app/oracle/oradata/ORCL/users01.dbf	2104850	ONLINE

- 최근에 받은 백업 데이터파일을 찾아 restore

SYS@orcl> ! cp -v backup/noarch/undotbs01.dbf /u01/app/oracle/oradata/ORCL/
'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

- 백업 이후에 변경 정보(리두)를 적용

SYS@orcl> recover datafile 4;

Media recovery complete.

SYS@orcl> alter database open;

Database altered.

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a, v\$tablespace b

where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2304883	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2304883	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2304883	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2304883	ONLINE

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil undo_datafile_num, t.ubablk undo_block, t.used_ublk from v\$session s, v\$transaction t, v\$rollname r where s.taddr = t.addr and t.xidusn = r.usn;

no rows selected

SYS@orcl> SYS@orcl> select salary from hr.employees where employee_id = 100;

```
SALARY
-----
30000
```

- Redo 적용되어 나옴
- TX이 걸렸던 update된 데이터는 커밋 안해서 자동 롤백이 될 수도 있음.(리커버리시점에서 진행)

<<시나리오8>> undo datafile 손상(백업 이후 리두정보 X)

- 언제나 백업해도 지나치지 않다.

```
[oracle@oel7v9r1 noarch]$ cp -v /u01/app/oracle/oradata/ORCL/*.* /home/oracle/backup/noarch/
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> './control01.ctl'
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> './control02.ctl'
'/u01/app/oracle/oradata/ORCL/redo01.log' -> './redo01.log'
'/u01/app/oracle/oradata/ORCL/redo02.log' -> './redo02.log'
'/u01/app/oracle/oradata/ORCL/redo03.log' -> './redo03.log'
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> './sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> './system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> './temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> './undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> './users01.dbf'
```

SYS@orcl> startup

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a, v\$tablespace b
where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2306036	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2306036	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2306036	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2306036	ONLINE

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

```
SYS@orcl> select s.username, s.sid, s.serial#, r.name, t.ubafil, t.xidusn, t.ubablk, t.used_ublk from v$session s, v$transaction t,
v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

no rows selected

<<hr>>

```
SYS@orcl> SYS@orcl> select salary from hr.employees where employee_id = 100;
```

SALARY

30000

```
SYS@orcl> update hr.employees set salary = salary * 1.1 where employee_id = 100;
```

1 row updated.

```
SYS@orcl> select salary from hr.employees where employee_id = 100;
```

SALARY

33000

<<sys>>

```
SYS@orcl> select s.username, s.sid, s.serial#, r.name, t.ubafil, t.xidusn, t.ubablk, t.used_ublk from v$session s, v$transaction t,
v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

USERNAME	SID	SERIAL#
-----		-----
NAME	UBAFIL	XIDUSN
-----	UBABLK	USED_UBLK

SYS	1	49282
_SYSSMU6_813816332\$	4	6
	214	1

kill 시켜야 할 경우(만약 킬 될 경우 자동롤백은 PMON이 진행해준다.)

instance recovery의 경우 SMON이 진행

```
alter system kill session '28.42766' immdiate;
```

리두데이터 로그스위치로 삭제

```
SYS@ora19c> alter system switch logfile;
```

System altered.

SYS@ora19c> /

System altered.

SYS@ora19c> /

System altered.

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory
```

```
SYS@ora19c> alter system checkpoint;
```

System altered.

```
-- 얼럿로그 Completed checkpoint up to RBA [0x111.5.10], SCN: 3268354
```

<<new session>>

```
HR@ora19c> update hr.employees set salary = salary * 1.1 where employee_id = 200;
```

1 row updated.

또는

```
[oracle@oel7v9r1 ~]$ sqlplus hr/hr
```

ERROR:

ORA-01116: error in opening database file 4

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-27041: unable to open file

Linux-x86_64 Error: 2: No such file or directory

Additional information: 3

<<sys>>

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil undo_datafile_num,
t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

USERNAME	SID	SERIAL#	UNDO_SEG_NUM	UNDO_SEG_NAME	UNDO_DATAFILE_NUM	UNDO_BLOCK	USED_UBLK
HR	20	48811	9	_SYSSMU9_2100413762\$	2		543
1							
HR	234	25867	8	_SYSSMU8_4078897598\$	2		323
1							

또는

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil undo_datafile_num,
t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr = t.addr and t.xidusn = r.usn;
```

USERNAME	SID	SERIAL#	UNDO_SEG_NUM	UNDO_SEG_NAME	UNDO_DATAFILE_NUM	UNDO_BLOCK	USED_UBLK
SYS	1	49282	6	_SYSSMU6_813816332\$	4	214	1

```
SYS@orcl> shut immediate
ORA-01116: error in opening database file 4
ORA-01110: data file 4: 'u01/app/oracle/oradata/ORCL/undotbs01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

```
SYS@orcl> shut immediate
ORA-01116: error in opening database file 4
ORA-01110: data file 4: 'u01/app/oracle/oradata/ORCL/undotbs01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
SYS@orcl> shut abort
ORACLE instance shut down.
```

```
SYS@orcl> startup
ORACLE instance started.
```

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes
 Variable Size 754974720 bytes
 Database Buffers 369098752 bytes
 Redo Buffers 7880704 bytes
 Database mounted.

ORA-01157: cannot identify/lock data file 4 - see DBWR trace file

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME
CON_ID				
4	ONLINE	ONLINE FILE NOT FOUND	0	

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2307423	SYSTEM
3	SYS_AUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2307423	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2307423	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2307423	ONLINE

- 최근에 받은 백업 데이터파일을 찾아 restore

SYS@ora19c> ! cp -v backup/noarch/undotbs01.dbf /u01/app/oracle/oradata/ORA19C/
 'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORA19C/undotbs01.dbf'

- 백업 이후에 변경 정보(리두)를 이용해서 복구작업을 시도했지만 리두정보가 없어서 실패

SYS@orcl> recover tablespace undotbs1

ORA-00279: change 2306033 generated at 05/05/2025 18:10:59 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_100_1199933387.dbf

ORA-00280: change 2306033 for thread 1 is in sequence #100

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

AUTO

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_100_1199933387.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_100_1199933387.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' offline drop;
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2307423	SYSTEM
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2307423	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2307423	RECOVER
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2307423	ONLINE

```
SYS@orcl> alter database open;
```

Database altered.

```
SYS@ora19c> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1230671320\$	PUBLIC	UNDOTBS1	OFFLINE
2	_SYSSMU2_3151764486\$	PUBLIC	UNDOTBS1	OFFLINE
3	_SYSSMU3_1295854473\$	PUBLIC	UNDOTBS1	OFFLINE
4	_SYSSMU4_1947518621\$	PUBLIC	UNDOTBS1	OFFLINE
5	_SYSSMU5_1526312829\$	PUBLIC	UNDOTBS1	OFFLINE
6	_SYSSMU6_2680029761\$	PUBLIC	UNDOTBS1	OFFLINE
7	_SYSSMU7_2189229235\$	PUBLIC	UNDOTBS1	OFFLINE
8	_SYSSMU8_4078897598\$	PUBLIC	UNDOTBS1	OFFLINE
9	_SYSSMU9_2100413762\$	PUBLIC	UNDOTBS1	OFFLINE
10	_SYSSMU10_2605481461\$	PUBLIC	UNDOTBS1	OFFLINE

11 rows selected.

또는

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY

11 rows selected.

```
SYS@orcl> create undo tablespace undo1 datafile '/u01/app/oracle/oradata/ORCL/undo01.dbf' size 100m autoextend on;
```

Tablespace created.

```
SYS@orcl> show parameter undo_tablespace
```

NAME	TYPE	VALUE
undo_tablespace	string	UNDOTBS1

```
SYS@orcl> alter system set undo_tablespace = undo1;
```

System altered.

```
SYS@orcl> show parameter undo_tablespace
```

NAME	TYPE	VALUE
undo_tablespace	string	UNDO1

```
SYS@ora19c> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1230671320\$	PUBLIC	UNDOTBS1	OFFLINE
2	_SYSSMU2_3151764486\$	PUBLIC	UNDOTBS1	OFFLINE
3	_SYSSMU3_1295854473\$	PUBLIC	UNDOTBS1	OFFLINE
4	_SYSSMU4_1947518621\$	PUBLIC	UNDOTBS1	OFFLINE
5	_SYSSMU5_1526312829\$	PUBLIC	UNDOTBS1	OFFLINE
6	_SYSSMU6_2680029761\$	PUBLIC	UNDOTBS1	OFFLINE

7_SYSSMU7_2189229235\$	PUBLIC	UNDOTBS1	OFFLINE
8_SYSSMU8_4078897598\$	PUBLIC	UNDOTBS1	OFFLINE
9_SYSSMU9_2100413762\$	PUBLIC	UNDOTBS1	OFFLINE
10_SYSSMU10_2605481461\$	PUBLIC	UNDOTBS1	OFFLINE
11_SYSSMU11_4210166392\$	PUBLIC	UNDO1	ONLINE
12_SYSSMU12_2672012841\$	PUBLIC	UNDO1	ONLINE
13_SYSSMU13_150914669\$	PUBLIC	UNDO1	ONLINE
14_SYSSMU14_2738526576\$	PUBLIC	UNDO1	ONLINE
15_SYSSMU15_2078482011\$	PUBLIC	UNDO1	ONLINE
16_SYSSMU16_2486507882\$	PUBLIC	UNDO1	ONLINE
17_SYSSMU17_3451577304\$	PUBLIC	UNDO1	ONLINE
18_SYSSMU18_833442847\$	PUBLIC	UNDO1	ONLINE
19_SYSSMU19_2109334265\$	PUBLIC	UNDO1	ONLINE
20_SYSSMU20_1507558699\$	PUBLIC	UNDO1	ONLINE

21 rows selected.

또는

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS

0	SYSTEM	SYS	SYSTEM	ONLINE
1_SYSSMU1_1261223759\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
2_SYSSMU2_27624015\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
3_SYSSMU3_2421748942\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
4_SYSSMU4_625702278\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
5_SYSSMU5_2101348960\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
6_SYSSMU6_813816332\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
7_SYSSMU7_2329891355\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
8_SYSSMU8_399776867\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
9_SYSSMU9_1692468413\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY
10_SYSSMU10_930580995\$		PUBLIC	UNDOTBS1	NEEDS RECOVERY

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS

11_SYSSMU11_1108405354\$		PUBLIC	UNDO1	ONLINE
12_SYSSMU12_765599853\$		PUBLIC	UNDO1	ONLINE
13_SYSSMU13_3057593607\$		PUBLIC	UNDO1	ONLINE
14_SYSSMU14_1232475546\$		PUBLIC	UNDO1	ONLINE
15_SYSSMU15_2798485940\$		PUBLIC	UNDO1	ONLINE
16_SYSSMU16_1082783005\$		PUBLIC	UNDO1	ONLINE
17_SYSSMU17_3127230972\$		PUBLIC	UNDO1	ONLINE
18_SYSSMU18_2934560322\$		PUBLIC	UNDO1	ONLINE
19_SYSSMU19_3811326955\$		PUBLIC	UNDO1	ONLINE

20 _SYSSMU20_2391515319\$ PUBLIC UNDO1 ONLINE

21 rows selected.

```
SYS@orcl> drop tablespace undotbs1 including contents and datafiles;
```

```
drop tablespace undotbs1 including contents and datafiles
```

*

ERROR at line 1:

ORA-01548: active rollback segment '_SYSSMU1_1261223759\$' found, terminate dropping tablespace

또는

```
SYS@ora19c> drop tablespace undotbs1 including contents and datafiles;
```

Tablespace dropped.

- 트랜잭션이 걸렸던 update된 데이터는 커밋 안해서 자동 롤백.

아니면,

-수동으로 needs recovery가 되어 있는 테이블스페이스를 offline으로 변환.

<Recovery>

```
SYS@orcl> create pfile from spfile;
```

File created.

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

```
SYS@orcl> !vi $ORACLE_HOME/dbs/initiorcl.ora → 아래 내용 추가
```

```
_offline_rollback_segments=(
```

```
_SYSSMU1_1261223759$,
```

```
_SYSSMU2_27624015$,
```

```
_SYSSMU3_2421748942$,
```

```
_SYSSMU4_625702278$,
```

```
_SYSSMU5_2101348960$,
```

```
_SYSSMU6_813816332$,
```

```
_SYSSMU7_2329891355$,
```

```
_SYSSMU8_399776867$,
```

```
_SYSSMU9_1692468413$,
```

```
_SYSSMU10_930580995$)
```

```
SYS@orcl> startup pfile=$ORACLE_HOME/dbs/initiorcl.ora
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

Database opened.

SYS@orcl> show parameter spfile

NAME	TYPE	VALUE

spfile	string	

SYS@orcl> drop tablespace undotbs1 including contents and datafiles;

Tablespace dropped.

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS

3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2409742	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2409742	SYSTEM
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2409742	ONLINE
5	UNDO1	/u01/app/oracle/oradata/ORCL/undo01.dbf	2409742	ONLINE

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS

0	SYSTEM	SYS	SYSTEM	ONLINE
11	_SYSSMU11_1108405354\$	PUBLIC	UNDO1	ONLINE
12	_SYSSMU12_765599853\$	PUBLIC	UNDO1	ONLINE
13	_SYSSMU13_3057593607\$	PUBLIC	UNDO1	ONLINE
14	_SYSSMU14_1232475546\$	PUBLIC	UNDO1	ONLINE
15	_SYSSMU15_2798485940\$	PUBLIC	UNDO1	ONLINE
16	_SYSSMU16_1082783005\$	PUBLIC	UNDO1	ONLINE
17	_SYSSMU17_3127230972\$	PUBLIC	UNDO1	ONLINE
18	_SYSSMU18_2934560322\$	PUBLIC	UNDO1	ONLINE
19	_SYSSMU19_3811326955\$	PUBLIC	UNDO1	ONLINE
20	_SYSSMU20_2391515319\$	PUBLIC	UNDO1	ONLINE

11 rows selected.

<원래대로 돌리기 1>

```
SYS@orcl> shut immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

```
SYS@orcl> startup          → spfile로 시작하기
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

Database opened.

```
SYS@orcl> show parameter undo_tablespace
```

NAME	TYPE	VALUE
undo_tablespace	string	UNDO1

```
SYS@orcl> show parameter spfile
```

NAME	TYPE	VALUE
spfile	string	/u01/app/oracle/product/19.3.0 /dbhome_1/dbs/spfileorcl.ora

```
SYS@orcl> alter system set undo_tablespace = undotbs1 scope=spfile;
```

System altered.

```
SYS@orcl> shut abort
```

ORACLE instance shut down.

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.*
```

```
SYS@orcl> ! cp -v /home/oracle/backup/noarch/* /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'/home/oracle/backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
'/home/oracle/backup/noarch/initiorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initiorcl.ora'
'/home/oracle/backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'/home/oracle/backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
'/home/oracle/backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
'/home/oracle/backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
```

```
'/home/oracle/backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 754974720 bytes
```

```
Database Buffers 369098752 bytes
```

```
Redo Buffers 7880704 bytes
```

```
Database mounted.
```

```
Database opened.
```

```
SYS@orcl> show parameter undo_tablespace
```

NAME	TYPE	VALUE
undo_tablespace	string	UNDOTBS1

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

```
11 rows selected.
```

```
<원래대로 돌리기 2>
```

```
SYS@orcl> create pfile from spfile;
```

```
File created.
```

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> ! cp -v backup/noarch/* /u01/app/oracle/oradata/ORCL/
```

```
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'backup/noarch/initiorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initiorcl.ora'
'backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
'backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

SYS@orcl> !

[oracle@oel7v9r1 ~]\$ cd \$ORACLE_HOME/dbs

[oracle@oel7v9r1 dbs]\$ vi initiorcl.ora

```
-- *.undo_tablespace='UNDO1' -> *.undo_tablespace='UNDOTBS1' 변경 필요(vi편집기 G 맨마지막으로)
-- pfile을 수정
```

SYS@orcl> startup pfile=\$ORACLE_HOME/dbs/initiorcl.ora

-- pfile로 시작

ORACLE instance started.

Database mounted.

Database opened.

SYS@orcl> create spfile from pfile;

File created.

[oracle@oel7v9r1 ~]\$ cd /u01/app/oracle/oradata/ORCL/

[oracle@oel7v9r1 ~]\$ ls

control01.ctl redo01.log redo03.log system01.dbf undo01.dbf users01.dbf

initiorcl.ora redo02.log sysaux01.dbf temp01.dbf undotbs01.dbf

[oracle@oel7v9r1 ~]\$ rm -rf undo01.dbf

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE

10 _SYSSMU10_930580995\$

PUBLIC UNDOTBS1

ONLINE

11 rows selected.

- 기존 생성했던 undo01.dbf 파일이 그대로 있으므로 해당 파일 삭제
- 있어도 상관은 없으나 새로 생성시 동일 이름 생성 불가